The Gap It Closed

Student copy (project or hand out)

1 NASA launched Artemis II from Launch Pad 39B at the Kennedy Space Center on April 1, 2026, at 6:35 in the evening, Eastern time. The number that mattered was not the date. No crew had flown beyond low Earth orbit since Apollo 17 came home in December 1972, a gap of fifty-four years, longer than most of the careers of the engineers who closed it.

2 The rocket was the heaviest part of the answer. The Space Launch System in its Block 1 form stands 322 feet tall and leaves the pad on 8.8 million pounds of thrust, about fifteen percent more than the Saturn V that carried the last crews; its core stage burns four RS-25 engines, each rated at 512,000 pounds of thrust in vacuum, while two solid rocket boosters supply three quarters of the liftoff force and then fall away into the Atlantic. The boosters do most of the early work. Then they are gone.

3 On top of it rode Orion, the capsule the crew named Integrity. Its crew module holds 330 cubic feet of habitable space, fifty-seven percent more than the command module that first carried astronauts to the Moon, enough to keep four people alive for twenty-one days undocked; its heat shield, 16.5 feet across, wears Avcoat reformulated from the Apollo capsules, built to char away in layers while the air outside climbs past 3,000 degrees Celsius. Much of what keeps Orion running is not American. The European Service Module, built by Airbus in Bremen with parts from thirteen countries, feeds the craft its power, its water, its oxygen and nitrogen, its heat control, and its push through space.

4 The crew was four. Reid Wiseman commanded; Victor Glover flew as pilot; Christina Koch and Jeremy Hansen rode as mission specialists. Koch became the first woman ever to travel beyond low Earth orbit, and Hansen, a Canadian, the first person who was not a United States citizen to go that far. They flew a free-return path, the same gravity-guided loop that brought Apollo 13 home, so that the pull of two worlds did most of the steering.

5 On April 6 the spacecraft reached 252,756 miles from Earth, passing Apollo 13 old record by 4,101 miles, the farthest from home any human beings have ever been. They came within 4,067 miles of the lunar surface, saw terrain on the far side no human eye had directly seen, and watched a total eclipse from space last fifty-seven minutes. Nine days after launch the crew splashed down in the Pacific off San Diego. The rocket and the capsule had each now carried people for the first time. Whether the whole architecture works is a question the next flight will answer.

AP-style multiple choice:

1. In the second paragraph, the writer follows the long technical sentence about the Space Launch System with the short sentence "Then they are gone" primarily to

A) correct an exaggeration in the preceding sentence about how much thrust the boosters supply

B) introduce a note of regret about the hardware that is discarded during launch

C) signal that the rocket's most important systems have now all been named

D) mirror in the rhythm of the prose the sudden departure of the boosters the sentence describes

2. Which of the following best characterizes the writer's tone in the final paragraph?

A) Wistful, dwelling on the dangers the crew faced and the uncertainty of their return

B) Measured and admiring, presenting extraordinary results plainly while reserving judgment on what they prove

C) Triumphant, treating the record distance and the eclipse as proof that the program has succeeded

D) Skeptical, implying that the mission's achievements may have been overstated by those who report them

Discussion questions:

1. The piece never tells the reader that Artemis II was amazing; it piles up numbers and named parts instead. Pick one paragraph and explain how the accumulation of concrete detail produces a sense of scale or wonder that an adjective like 'incredible' would not.

2. The final sentence withholds a verdict, saying only that the next flight will answer whether the whole architecture works. How does ending on an admitted uncertainty, rather than on the distance record, shape your trust in the writer?

Writing prompt (Homework or extended in-class, Q2 rhetorical analysis):

Read the opener carefully. Then write an essay analyzing the rhetorical choices the writer makes to convey the scale and significance of Artemis II without stating outright that the mission is remarkable. In your response, consider how choices such as sentence length and variation, the accumulation of concrete technical detail, the contrast between long sentences and short ones, and the writer's candor about what remains unknown work together to build the piece's effect.